

Anonymity Preserving, Decoupled Distributed Authentication for Multihop IoT Networks

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Abstract—The Internet of Things (IoT) demands lightweight and secure authentication for resource constrained multihop networks, while also protecting the privacy of individual devices. Many distributed authentication schemes have been proposed in the past for IoT networks. Among those schemes, the decoupled distributed authentication scheme (DAuth) has been proposed recently for multihop IoT networks. However, we have identified two critical gaps in that scheme. First, the real device identity is exposed on the open channel in every authentication session, enabling adversarial cross-session tracking. Second, packet loss in the session key establishment phase desynchronises the device and server states, forcing full re-enrolment. Therefore, this paper proposes a security improved and anonymity preserving decoupled distributed authentication scheme, addressing both the identified gaps. Device anonymity is preserved through per-session pseudonym rotation, where a rotating pseudonym replaces the real device identity in all open-channel communications, preventing long-term tracking. A dual-state synchronisation mechanism maintained at both the authentication server and the gateway/root enables seamless recovery from single packet-loss events without re-enrolment. The proposed scheme is evaluated via ProVerif-based formal security analysis, and theoretical and simulation-based performance analysis confirm that the proposed scheme reduces energy by 42.8 % and 34.6 %, and CPU time by 42.7 % and 35.3 %, over the state-of-the-art LAAKA and Zhou schemes, respectively, while maintaining the lowest overhead among all compared schemes.

Index Terms—IoT Authentication, Distributed Authentication, Anonymity, Desynchronization in Authentication, Dual-State Storage.

I. INTRODUCTION

The Internet of Things (IoT) connects millions of smart devices that collect and transmit data to centralized platforms [1] such as gateways, often through multihop paths [2]. Industrial and outdoor deployments expose devices to a range of adversarial threats, making authentication a prerequisite for any communication session. Because IoT nodes are battery-powered and resource-constrained, authentication schemes must be lightweight as well as secure [3].

Centralized schemes route all authentication traffic through a gateway, imposing excessive relay load on intermediate nodes in a multihop topology [4] [5] [6]. Distributed schemes distribute this responsibility [7]. However, a plain parent centric distributed authentication schemes overload whichever node a set of new devices happened to join under, while all other capable nodes remain idle [7].

DAuth [8] addressed this skewed-load problem by proposing a lightweight decoupled distributed authentication scheme in which the gateway or root node pre-assigns any capable IoT node (parent or non-parent) as the authentication server for newly joined devices. By decoupling authentication from network joining process, the scheme distributes the authentication load evenly across the network. Authentication process relies on Physically Unclonable Function (PUF)-based secret derivation, a hash-based one-way accumulator and XOR-masked message exchanges, resulting in an efficient and lightweight authentication protocol.

Despite its efficiency, DAuth [8] leaves two security gaps unaddressed. First, the real device identity is transmitted in plaintext over the open channel in every authentication session, enabling an adversary with radio access to track a device across sessions, a direct violation of modern privacy requirements. Second, both the authentication server and the device maintain only a single session state and so a packet loss during session key delivery desynchronises their states, causing all subsequent authentication attempts to fail until the device undergoes full re-enrolment, which is an energy-costly operation for battery-powered constrained nodes.

This paper addresses both gaps with two lightweight extensions to DAuth. To close the anonymity gap, the real identity is never placed on the open channel after enrolment, it is replaced by a rotating pseudonym that is refreshed after every key exchange, so successive sessions become unlinkable to an observer. To eliminate the re-enrolment penalty caused by desynchronisation, both the authentication server and the gateway maintain a dual state, if the session-key update is lost in transit, the device still arrives under its previous pseudonym, which the server recognises and uses to re-synchronise without repeating enrolment.

This paper makes the following key contributions.

- 1) A anonymity preserving scheme is proposed that introduces per-session pseudonym rotation to preserve device anonymity and a dual-state mechanism to recover from desynchronisation without re-enrolment.
- 2) The security of the proposed scheme is validated through both informal analysis and formal verification using ProVerif.
- 3) The efficiency of the proposed scheme is evaluated using both theoretical and simulation-based (COOJA/Contiki-

NG) frameworks.

Section II elaborates on related literature. Section III presents the proposed scheme. Furthermore, Section IV contains the detailed security analysis. Finally, Section V presents the performance analysis, with Section VI stating the conclusion.

II. LITERATURE SURVEY

Existing schemes for IoT authentication address the problem from multiple directions but each leaves gaps. Distributed approaches reduce relay load but remain topology-bound. LightSAE [7] authorise parent nodes as authentication servers, yet a popular parent is overwhelmed by simultaneous join requests from many children. Zheng et al. [9] designed a peer-to-peer PUF-based scheme that lacks a multihop decoupled design. RAAF-MEC [10] deploy multiple PUF-equipped edge servers, but the absence of a multihop topology means certain servers still receive disproportionate load. PUF-based schemes avoid long-term secret storage but trade in overhead. Li et al. [11] and SAKMS [12] combine PUF with session key exchange at the cost of ECC operations, while He et al. [5] achieve lightweight anonymous authentication using hash-and-XOR primitives for CoAP/OSCORE but rely on a centralized server. DAAuth [8] removes this topology coupling by letting the gateway or root node assign any capable node, parent or not, as the authentication server, distributing the authentication load evenly across a genuine multihop network.

Identity-hiding approaches introduce anonymity at prohibitive cost. Al-Riyami and Paterson [13] propose ECC-based certificateless cryptography that is too heavy for constrained nodes, and blockchain-based schemes [14] achieve decentralisation but incur prohibitive storage and consensus overhead. Zhong et al. [15] present a systematic taxonomy of lightweight anonymous IoT authentication, deconstructing existing protocols into nine sub-frameworks (SF1-UA through SF9-FS) and two general frameworks (GF1, GF2), and their analysis demonstrates that only GF2 simultaneously achieves user anonymity, forward secrecy, and desynchronisation resistance, a combination absent from all prior lightweight schemes. The proposed scheme follows the GF2 architectural pattern, where the dual-state pseudonym mechanism corresponds to the SF6-UA sub-framework [15], and per-session key derivation from a PUF-assisted fresh nonce provides forward secrecy analogous to GF2. Unlike GF2, which uses a one-time hash chain $hk = h(hk)$, the proposed scheme replaces the hash chain with PUF-derived session randomness and augments authentication with a hash-based accumulator — mechanisms not modelled in the Zhong et al. taxonomy. Table I contrasts key features across representative schemes. Existing lightweight schemes trade off at least one of these properties, namely decoupled design, PUF, accumulator-based authentication, authentication secret update (ASU), anonymity, or desynchronisation recovery, whereas the proposed scheme is the first to satisfy all six simultaneously with low computational and communication overhead.

TABLE I
FEATURES COMPARISON OF AUTHENTICATION SCHEMES.

Scheme	Dec.	PUF	Acc.	ASU	Anon.	Desync	C.OH	M.OH
RAAF-MEC [10]	×	✓	×	✓	×	×	H	L
SAKMS [12]	✓	×	×	×	×	×	H	L
Li et al. [11]	✓	✓	×	✓	×	×	H	L
Zheng et al. [9]	✓	✓	×	✓	×	×	M	L
DAAuth [8]	✓	✓	✓	✓	×	×	L	L
Proposed	✓	✓	✓	✓	✓	✓	L	L

Dec.: Decoupled design. *Acc.*: Hash-based accumulator. *ASU*: Auth. secret update. *Anon.*: Device anonymity via pseudonym. *Desync*: Desync. recovery. *C.OH*: Computation — L (Hash/XOR/≤ 2 AES), M (> 2 AES), H (ECC/Pairings). *M.OH*: Message exchanges — L (1–3), M (4–5), H (> 5).

III. PROPOSED AUTHENTICATION SCHEME

A. System Model and Assumptions

The proposed scheme is designed to help nodes establish secure multihop communications with the gateway. To establish a secure communication, nodes should be authenticated by a trusted authentication server. A set of existing IoT nodes inside the network is entrusted with the responsibility of authentication by the gateway in a distributed manner. The gateway also shares a secret key with each of those responsible IoT nodes/authentication servers. The authentication server can be a parent or a non-parent node, which may be *decoupled*, preventing newly joined child nodes from overwhelming their parent with biased authentication load.

IoT nodes are assumed to be embedded with a PUF and are capable of performing various cryptographic operations, such as encryption/decryption, XOR, hashing, bitwise AND, and concatenation.

Nodes D and AS are assumed to be the newly joined node and its authentication server, respectively. GW represents the gateway symbolically. AS may be located at a singlehop or multihop distance from D . It is assumed that GW has already selected the authentication server set and has established a shared secret key with each member of the set. Let $\mathcal{AS}' = \{AS_1, AS_2, \dots, AS_n\}$ be the set of authentication servers selected by GW , such that $AS \in \mathcal{AS}'$. GW has a shared secret key K_{GW-AS} with AS . It is also assumed that D , whose distinct identifier is ID_D , possesses the address of its authentication server AS along with its unique identity ID_{AS} . Table II elaborates symbols and functions incorporated in the following scheme.

There are mainly three phases: node enrolment, node authentication, and node session key exchange. The following sections elaborate on these phases.

B. Node Enrolment Phase

The node enrolment phase is the first phase undergone by a newly joined node D with its authentication server AS . Before entering this phase, it is assumed that D has the required information regarding AS : the address and unique identity (ID_{AS}) of AS . AS can lie at a single-hop or multi-hop distance from D . This phase is assumed to be executed in a secure environment by establishing an end-to-end encrypted channel between D and AS . Every newly joined node executes

TABLE II
NOTATION.

Symbol	Description
D, AS	Newly joined device and its authentication server
GW	Gateway / root node
$H(\cdot)$	Collision-resistant hash function (SHA-256)
$SE(\cdot), SD(\cdot)$	Symmetric encryption/decryption (AES-128)
$\oplus$	Bitwise XOR operation
$\&$	Bitwise AND operation
c_D, c_{AS-D}	PUF challenges for D (issued by AS) and AS (by D)
m_D	Session-based random issued by AS during enrolment
m_{curr}, m_{old}	Current and previous session-based randoms at D and AS
y_D, R_D	Unique secret and PUF response of D
Y_D^H	Hashed unique secret: $H(y_D)$
T_{Acc}	Accumulated authentication value at AS
Φ_{AS-D}	PUF-response binding: $PUF_{AS}(c_{AS-D}) \oplus R_D$
PID_{curr}, PID_{old}	Current and previous pseudonyms of D
K_{GW-D}	Session key between GW and D
K_{GW-AS}	Long-term shared key between GW and AS
ts_1, ts_2, ts_{auth}	Timestamps for freshness verification

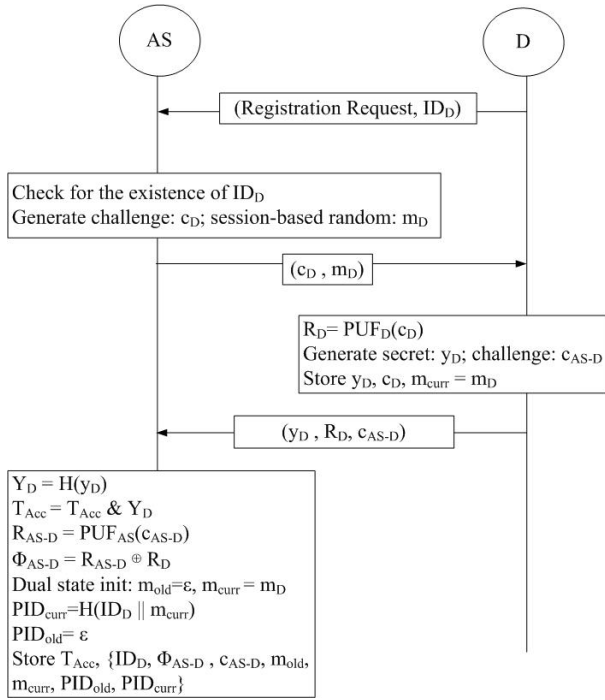


Fig. 1. Enrolment Phase via secure channel.

this phase once until it disconnects from the network. The main purpose of this phase is to enrol the unique secret of D (i.e., y_D) with the accumulated value (T_{Acc}) of AS . During the node authentication phase, D uses y_D again to authenticate with AS . PUF responses and a session-based random are also communicated between AS and D during this phase to establish the secure communication of y_D during authentication. Figure 1 outlines the methodology of this phase.

D initiates the phase by sending a registration request along with its unique identity ID_D to AS . As a response, AS generates a challenge c_D from the predefined challenge set $\mathbb{C}_D$ and a session-based random m_D ; these are then forwarded securely to D . Both of these credentials act to secure the

communication during authentication: c_D enables D to regenerate its PUF response, while m_D introduces unpredictability into subsequent open-channel messages, preventing adversarial prediction. With c_D as input to its embedded PUF_D , D obtains a unique response $R_D = PUF_D(c_D)$. After this, D generates its own unique secret value y_D , which is also a random number; the security of the entire scheme critically depends on maintaining the confidentiality of y_D . D also picks a reverse challenge c_{AS-D} from the predefined challenge set $\mathbb{C}_{AS-D}$ and sends (y_D, R_D, c_{AS-D}) back to AS through the secure channel, maintaining the confidentiality of y_D . D then stores $\{y_D, c_D, m_D\}$. Unlike DAuth [8], which stores m_D as a plain session random, the proposed scheme retains it as the initial *current* session random m_{curr} , seeding the pseudonym mechanism described below.

AS inputs the secret y_D to the hash function $H(\cdot)$, generating a tag value $Y_D^H = H(y_D)$. After that, AS performs a bitwise AND operation between the calculated tag and the present accumulated value T_{Acc} , obtaining the updated $T_{Acc} = T_{Acc} \& Y_D^H$. The accumulated value contains the compact, hashed, unique secrets of individual nodes authenticated by AS . In this way, D successfully enrolls itself with AS . Further, AS generates a unique response $R_{AS-D} = PUF_{AS}(c_{AS-D})$ using the received challenge c_{AS-D} in the embedded PUF_{AS} , and calculates $\Phi_{AS-D} = R_{AS-D} \oplus R_D$ by operating XOR between R_{AS-D} and the received R_D . In this way, R_D is kept in Xored form for further secure communications by AS . At the end, AS creates an entry against ID_D and stores the calculated Φ_{AS-D} along with the challenge c_{AS-D} and the received session-based random m_D .

The proposed scheme extends this base enrolment with a pseudonym and dual-state initialisation to eliminate ID_D from all future open-channel messages. Since ID_D is transmitted only over this secure enrolment channel and must never appear on the open channel again, D will henceforth be identified by the rotating *pseudonym* $PID = H(ID_D || m_{curr})$ in all future authentication messages. Because H is one-way and m_{curr} is a private session random, successive pseudonyms are computationally unlinkable by a channel observer. AS pre-computes $PID_{curr} = H(ID_D || m_{curr})$ and sets $PID_{old} = \perp$, $m_{old} = \perp$, forming the initial *dual state*. This dual state ($m_{curr}, m_{old}, PID_{curr}, PID_{old}$) is maintained to handle desynchronisation: if the updated m_{new} delivered in the Key Exchange Phase is lost in transit, AS can still recognise D by its previous pseudonym and restore synchronisation without requiring re-enrolment. AS therefore stores $\{T_{Acc}, ID_D, \Phi_{AS-D}, c_{AS-D}, m_{curr}, m_{old}, PID_{curr}, PID_{old}\}$.

C. Node Authentication Phase

The node authentication phase is the second phase of the proposed scheme. Its main purpose is to authenticate D 's unique secret y_D with AS . Since this phase is executed over an open channel, y_D must be communicated in secure form, preventing it from being intercepted by adversaries. This security is established through PUF-based challenge-response, producing a secret key R_D , and the session-based random

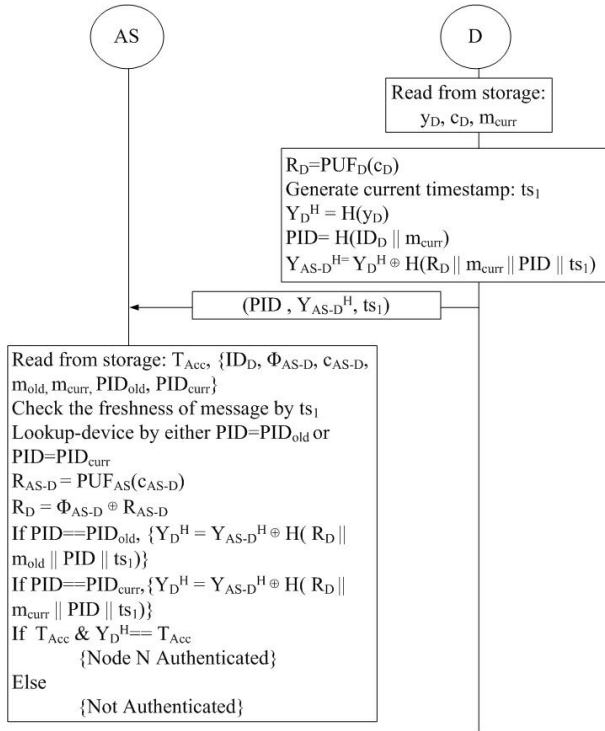


Fig. 2. Authentication Phase via open channel.

m_{curr} incorporating unpredictability. Both R_D and m_{curr} are known only to D and AS . Figure 2 illustrates the phase.

The phase is initiated by D . D regenerates $R_D = \text{PUF}_D(c_D)$ from the stored challenge c_D obtained during enrolment. D then computes its current pseudonym $PID = H(ID_D || m_{curr})$, which is used in place of the real identity ID_D in the outgoing message, unlike DAAuth [8], which sends ID_D directly, enabling cross-session tracking. D produces the hashed secret $Y_D^H = H(y_D)$ in XORed form as $Y_{AS-D}^H = H(y_D) \oplus H(R_D || m_{curr} || PID || ts_1)$. The XOR mask is the hash of R_D regenerated from challenge c_D , the session-based random m_{curr} , the pseudonym PID , and the current timestamp ts_1 . The timestamp is used to provide replay protection to the XORed value Y_{AS-D}^H . The complete message (PID, Y_{AS-D}^H, ts_1) is then sent to AS . An attacker will not be able to obtain $H(y_D)$ from Y_{AS-D}^H without knowing R_D (which can only be generated by D 's PUF) and m_{curr} (known only to D and AS through enrolment).

After receiving the message, AS checks for the existence of an entry matching the received PID . In a departure from DAAuth [8], AS performs a dual-state lookup: it checks whether the received PID matches PID_{curr} or PID_{old} in its records. A match on PID_{old} indicates that D missed the Key Exchange Phase delivery in the previous session and still holds the prior m_{curr} ; in this case, AS uses the stored m_{old} for the subsequent verification, recovering from the desynchronisation without requiring re-enrolment. AS verifies the freshness of ts_1 . R'_D is calculated as $\Phi_{AS-D} \oplus \text{PUF}_{AS}(c_{AS-D})$, where both Φ_{AS-D} and c_{AS-D} are retrieved from D 's enrolment record.

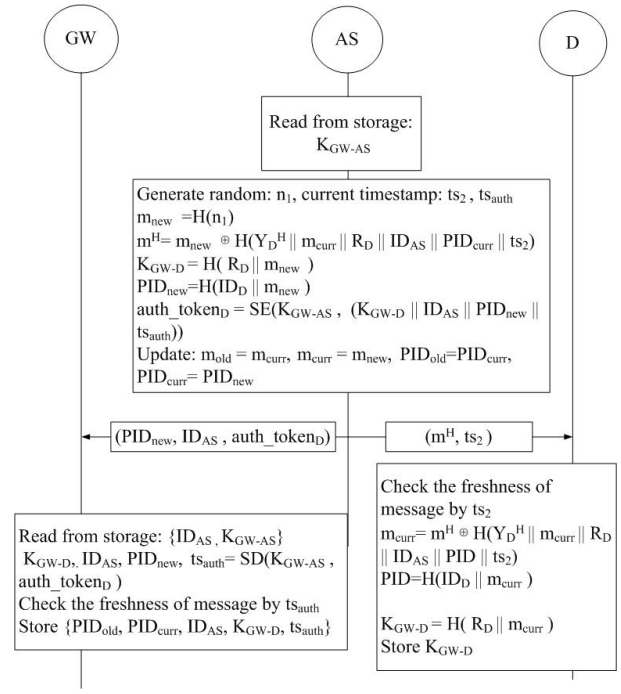


Fig. 3. Session Key Exchange Phase via open channel.

AS regenerates the same XOR mask and un.masks $Y_D^{H'}$, then performs the bitwise AND check: $T_{Acc}^{new} = T_{Acc} \& Y_D^{H'} \stackrel{?}{=} T_{Acc}$. From the properties of the hash-based one-way accumulator, if an element is combined into an accumulated value using bitwise AND and the same operation is again performed between them, it gives back the same accumulated value. If $T_{Acc}^{new} = T_{Acc}$, it proves that the unique secret of D is already a legitimate member of AS . With the successful operation, the authentication of D is completed by AS , and AS proceeds to the Key Exchange Phase.

D. Node Session Key Exchange Phase

This phase is executed by AS to inform GW about the authentication of the newly joined node D at the end of the node authentication phase. AS uses an auth_token_D to deliver the session key K_{GW-D} to GW , facilitating session key generation between GW and D . The proposed scheme extends DAAuth [8] by additionally deriving a fresh pseudonym $PID_{new} = H(ID_D || m_{new})$ at the end of each session and carrying it within auth_token_D , so that GW can update its pseudonym mapping for D and successive sessions remain unlinkable. Figure 3 shows the phase.

K_{GW-D} is calculated by AS using the hashed output of R'_D and m_{new} , i.e., $K_{GW-D} = H(R'_D || m_{new})$. R'_D is the PUF response of D regenerated by AS during the node authentication phase, and thus known only to AS and D . m_{new} , on the other hand, is computed by hashing a fresh random number n_1 generated by AS as $m_{new} = H(n_1)$. The computed m_{new} will also act as the updated session-based random for the next authentication session between AS and D , incorporating an update to the authentication secret. Further, AS derives

$PID_{new} = H(ID_D \| m_{new})$. Because PID_{new} is based on the fresh m_{new} , it is computationally unlinkable to the pseudonym used in the current session. The session key K_{GW-D} , the identities ID_{AS} and ID_D , the next pseudonym PID_{new} , and the authentication completion timestamp ts_{auth} are encrypted under the shared key K_{GW-AS} (retrieved from storage) to form $auth_token_D$. AS simultaneously sends (m_H, ts_2) to D and $(PID_{new}, ID_{AS}, auth_token_D)$ to GW . It then saves the current nonce and pseudonym as the previous values and advances its records: $m_{curr} \leftarrow m_{new}$, $PID_{curr} \leftarrow PID_{new}$, with m_{old} and PID_{old} retaining the superseded values.

On the other hand, AS also informs D of m_{new} so that D can independently compute the session key K_{GW-D} and its next pseudonym. Since m_{new} must be protected from adversaries (its capture would allow prediction of future session key material), AS masks it as $m_H = m_{new} \oplus H(Y_D^{H'} \| m_{curr} \| R'_D \| ID_{AS} \| PID_{curr} \| ts_2)$. Here, $Y_D^{H'}$ is the hashed form of the unique secret of D , known to AS from the authentication phase. Both m_{curr} and R'_D are likewise known to AS from the authentication phase. The current timestamp ts_2 is used to prevent replay of m_H in a future session.

Obtaining (m_H, ts_2) , D first verifies the freshness of ts_2 . D then recovers m_{new} by regenerating the hash mask using its locally held Y_D^H , R_D , m_{curr} , ID_{AS} , PID_{curr} , and the received ts_2 , and XORing with m_H . The input of the regenerated hash is already known by D . In this way, D and AS update their shared session-based random m_{curr} with m_{new} . Alongside, D establishes the session key $K_{GW-D} = H(R_D \| m_{new})$ and rotates its pseudonym to $PID_{curr} = H(ID_D \| m_{new})$, thus preparing itself for secure communication with GW .

GW decrypts the token with K_{GW-AS} , verifies ts_{auth} , and extracts K_{GW-D} and PID_{new} . GW saves PID_{curr} as the previous pseudonym, sets PID_{new} as the current one, and stores the session key against both. Retaining PID_{old} ensures that if the delivery (m_H, ts_2) to D was lost, GW can still service D 's data frames under the prior session key pending re-authentication.

E. Desynchronisation Recovery

During the Key Exchange Phase, AS simultaneously dispatches two packets: $(PID_{new}, ID_{AS}, auth_token_D)$ to GW and (m_H, ts_2) to D . Loss of either packet leads to a different desynchronisation scenario, each handled without re-enrolment.

If the $AS \rightarrow GW$ packet is lost, D has already received (m_H, ts_2) and updated its state to PID_{new} and K_{GW-D} . GW , however, never received PID_{new} or the session key, so it rejects D 's subsequent data frames. D detects the failure and re-initiates Phase 2 (Authentication) with its current PID_{curr} . Since D and AS are already synchronised, AS recognises PID_{curr} on the normal path, completes authentication, and forwards a fresh $auth_token_D$ to GW , restoring normal operation in one additional round.

The primary desynchronisation scenario arises when the $AS \rightarrow D$ packet (m_H, ts_2) is lost in transit. AS has already advanced its internal state by updating m_{curr} to m_{new} and

Algorithm 1 Desynchronisation Recovery

Precondition: D holds stale $m_{curr} = m_{old}$; AS holds $(m_{curr}, m_{old}, PID_{curr}, PID_{old})$

Authentication Phase

- 1: $PID \leftarrow H(ID_D \| m_{curr}) \triangleright$ stale $m_{curr} \Rightarrow PID = PID_{old}$
- 2: $D \rightarrow AS: (PID, Y_{AS-D}^H, ts_1)$
- 3: **if** $PID = PID_{curr}$ **then**
- 4: AS verifies using m_{curr} $\triangleright$ normal path
- 5: **else if** $PID = PID_{old}$ **then**
- 6: AS verifies using m_{old} $\triangleright$ desync detected
- 7: **else**
- 8: AS rejects D
- 9: **end if**

Key Exchange Phase (desync path)

- 10: $n_1 \leftarrow \text{rand}()$
- 11: $m_{new} \leftarrow H(n_1)$
- 12: $m_H \leftarrow m_{new} \oplus H(Y_D^{H'} \| m_{old} \| R'_D \| ID_{AS} \| PID_{old} \| ts_2)$
 $\triangleright AS$ uses m_{old}, PID_{old}
- 13: $AS \rightarrow D: (m_H, ts_2)$
- 14: $m_{new} \leftarrow m_H \oplus H(Y_D^H \| m_{curr} \| R_D \| ID_{AS} \| PID \| ts_2)$ $\triangleright$
 D uses its own $m_{curr} = m_{old}$ — recovers correctly
- 15: $m_{old} \leftarrow m_{curr}; m_{curr} \leftarrow m_{new}; PID \leftarrow H(ID_D \| m_{curr})$

Result: D and AS re-synchronised; PID rotated

rotating PID_{curr} to PID_{new} , while D retains the stale m_{curr} . On the next authentication attempt, D computes its pseudonym from the stale nonce, producing a PID that no longer matches any record at AS . The XOR mask used to protect Y_{AS-D}^H is likewise constructed from the stale m_{curr} , so AS cannot reconstruct the correct mask and the accumulator check fails. Authentication is therefore rejected. In schemes without a recovery mechanism, this failure forces D to execute the full enrolment phase again, a costly operation for a battery-powered node that re-exposes ID_D on the secure channel and resets all session state.

The proposed dual-state mechanism avoids this re-enrolment cost. If the Key Exchange Phase delivery (m_H, ts_2) is lost, D retains its stale m_{curr} , so the pseudonym it computes in the next session equals PID_{old} from AS 's perspective. Algorithm 1 shows how the scheme adapts the Authentication and Key Exchange Phases to recover without re-enrolment. D 's behaviour is *identical* in both paths: it always uses its own m_{curr} and PID . Only AS adapts: the dual-state lookup detects which nonce D is on, then constructs m_H with the matching (m_{old}, PID_{old}) so that D 's symmetric decryption succeeds and the states converge. The dual-state mechanism tolerates a single consecutive Key Exchange Phase loss, but two or more consecutive losses exhaust both slots, requiring re-enrolment, a deliberate trade-off between storage overhead and recovery depth.

IV. SECURITY ANALYSIS

A. Formal Security Analysis

The proposed scheme is translated into ProVerif [16] queries under the Dolev–Yao adversary model [17], where the adversary has full control of the public channel and can intercept, replay, modify, and inject messages. Ten queries are verified and all are satisfied. Figure 4 shows the verification summary.

1) *Protocol Correspondence*: Five correspondence queries verify that the protocol steps happen in the correct order and that no adversary can skip or forge a step.

The first query checks enrolment integrity: a device can only be registered at the authentication server after it has explicitly initiated the enrolment process. This rules out any possibility of a device being enrolled without its own participation.

The second query checks the device side of the authentication phase: the device receives the server’s acknowledgement and the fresh timestamp ts_2 only after it first sent a valid authentication request. This prevents an adversary from replaying a stale acknowledgement to trick the device into advancing its state.

The third query checks the server side of the authentication phase: the authentication server completes its processing only after it received a structurally correct request from the device. A forged or replayed message cannot push the server into a completed state.

The fourth query binds the Authentication and Key Exchange phases: the PUF response R_D and the session seed m_{curr} that the device commits during authentication are proven to be exactly the same values used in the key exchange. This rules out an active adversary who might try to substitute credentials between phases.

The fifth query covers key exchange: the device recovers the new session seed m_{new} only after it legitimately sent the key-exchange request. No key material can be delivered to the device out of sequence.

2) *Secrecy and Forward Secrecy*: Two queries verify the secrecy of the session key $K_{GW-D} = H(R_D \| m_{\text{new}})$ from the perspective of the device and the gateway independently. Even though the adversary observes all traffic on the public channel, it cannot reconstruct K_{GW-D} because the PUF response R_D is never transmitted and m_{new} is always delivered XOR-masked.

A separate query confirms forward secrecy: the rotated session seed m_{new} itself remains secret. This means that even if an adversary somehow recovers a current session key, it gains nothing about future sessions, because each m_{new} is derived from an independently generated fresh nonce.

3) *Anonymity Preservation*: The most significant secrecy result concerns the real device identity ID_D . After enrolment, ID_D is never sent over the open channel again. Every subsequent message instead carries the pseudonym $PID = H(ID_D \| m_{\text{curr}})$. ProVerif formally confirms that no attacker, even one who intercepts and manipulates every single message on the network, can recover ID_D from the observed traffic.

This result supports device anonymity. However, anonymity is only meaningful if the pseudonym used in one session cannot be linked to the pseudonym used in the next.

The forward-secrecy query addresses this: because m_{new} is proven secret, consecutive pseudonyms $H(ID_D \| m_{\text{curr}})$ and $H(ID_D \| m_{\text{new}})$ are computationally indistinguishable to any observer, making it difficult for an adversary who collects pseudonyms from multiple sessions to determine whether they belong to the same physical device.

4) *Offline Guessing Resistance*: Finally, a weak-secrecy query confirms that the session key K_{GW-D} cannot be guessed from the public transcript even by an adversary with unlimited offline dictionary power. This closes the gap against brute-force and dictionary attacks that target the key directly from captured packets.

B. Informal Analysis

1) *Replay Attack Resistance*: The Enrolment Phase is executed over a secure channel and immune to replay. The Authentication and Key Exchange Phases embed timestamps ($ts_1, ts_2, ts_{\text{auth}}$) and the session-based random m_{curr} into the hash masks. A replayed message with a stale timestamp or stale m_{curr} will fail the freshness check at AS , D , or GW respectively.

2) *MITM Attack Resistance*: The Enrolment Phase is conducted over an encrypted channel, which precludes MITM. In the Authentication and Key Exchange Phases, the PUF response R_D (device-specific, unreproducible) and session random m_{curr} (private to D and AS) are embedded in every XOR mask. Without R_D and the current m_{curr} , no adversary can forge a valid masked tag Y_{AS-D}^H or unmask m_H .

3) *Device Anonymity*: ID_D is transmitted only once, over the secure enrolment channel, and is never exposed on the open network again. All open-channel messages use $PID = H(ID_D \| m_{\text{curr}})$, which is refreshed to $H(ID_D \| m_{\text{new}})$ after every key exchange. Because m_{new} is derived from a freshly generated nonce n_1 , successive pseudonyms are computationally unlinkable.

4) *Desynchronisation Recovery*: Packet loss can occur at three distinct points in the Authentication and Key Exchange Phases, each handled without requiring re-enrolment.

If the acknowledgement and fresh timestamp ts_2 sent by AS to D at the end of the authentication phase are lost in transit, D does not advance its internal state and retains m_{curr} and the current PID unchanged. Since AS has not yet performed any state update at this stage, both entities remain consistent. Upon timeout, D retransmits the authentication request with the same PID and ts_1 , and the exchange proceeds normally.

The primary desynchronisation scenario arises when the Key Exchange Phase response carrying the masked session seed m_H , sent from AS to D , is lost. In this case, AS has already advanced its state to m_{new} and rotated the pseudonym to PID_{new} , while D retains the previous m_{curr} . On the subsequent authentication attempt, D constructs its pseudonym from the stale m_{curr} , producing PID_{old} . AS recognises this via the dual-state lookup — storing both $(m_{\text{curr}}, PID_{\text{curr}})$ and $(m_{\text{old}}, PID_{\text{old}})$ — and resumes from the consistent prior state, restoring synchronisation without any re-enrolment overhead.

Verification summary:

Query inj-event(DeviceEnrolled(id_D_2,id_AS_3)) ==> inj-event(DeviceEnrollmentStarts(id_D_2,id_AS_3)) is **true**.

Query inj-event(DeviceAuthenticated(id_D_2,id_AS_3)) ==> inj-event(DeviceAuthenticationStarts(id_D_2,id_AS_3)) is **true**.

Query inj-event(AuthenticationServerEnds(id_AS_3,id_D_2)) ==> inj-event(AuthenticationServerStarts(id_AS_3,id_D_2)) is **true**.

Query inj-event(AuthenticationEndsFull(id_D_2,id_AS_3,R_D_2,m_D_2)) ==> inj-event(AuthenticationStartsFull(id_D_2,id_AS_3,R_D_2,m_D_2)) is **true**.

Query event(DeviceKeyExDone(id_D_2,id_AS_3)) ==> event(DeviceKeyExStarts(id_D_2,id_AS_3)) is **true**.

Query not attacker(SecretK_GW_D_Device[]) is **true**.

Query not attacker(SecretK_GW_D_GW[]) is **true**.

Query not attacker(SecretM_New[]) is **true**.

Query not attacker(SecretID_D[]) is **true**.

Weak secret SecretK_GW_D_Device is **true**.

Fig. 4. ProVerif Output.

If the token forwarded from AS to GW is lost, GW does not receive PID_{new} or the associated session key K_{GW-D} . Consequently, D 's subsequent data packet, carrying PID_{new} , is not recognised by GW . D re-initiates from the Authentication Phase; the dual-state storage at AS ensures that this re-authentication succeeds and a fresh token is forwarded to GW , restoring normal operation.

5) *Forward Secrecy*: $K_{GW-D} = H(R_D || m_{new})$. Compromise of a session key does not reveal previous session keys because m_{new} is independently generated from a fresh nonce and R_D is PUF-derived.

V. PERFORMANCE ANALYSIS

A. Theoretical Performance Analysis

1) *Computational Cost*: The computational cost is a crucial deterministic factor ensuring the security and efficiency of an authentication scheme. Pycrypto, a Python cryptographic library, is used in a Windows system to measure the computational cost of core cryptographic functions, such as SHA-256 (T_{hash}), AES Encryption/Decryption (T_{aes}), and Random generation (T_{rand}). The PUF operation cost (T_{puf}) is taken from prior benchmarks on the same platform [9].

To execute the node enrolment phase (Figure 1):

- AS computes two random generations (T_{rand}), one PUF operation (T_{puf}), and two hash operations (T_{hash}) for pseudonym initialisation.
- On the other hand, D executes two random generations (T_{rand}) and one PUF operation (T_{puf}).

Hence, the total computational cost incurred per node enrolment amounts to $2T_{puf} + 2T_{hash} + 4T_{rand}$. As shown in Table III, the proposed scheme achieves an enrolment cost of 0.18 ms. It competes well with Zhou [18] (0.13 ms) while additionally providing PUF-based secret binding and

pseudonym initialisation, both of which are absent from those schemes.

TABLE III
COMPUTATIONAL COST COMPARISON: ENROLMENT PHASE.

Scheme	Operations	Cost (ms)
LAACA [19]	$3T_{hash}$	0.11
Zhou [18]	$T_{puf} + 2T_{hash} + 3T_{rand}$	0.13
Zheng et al. [9]	$2T_{puf} + 8T_{hash} + 4T_{aes} + 5T_{rand}$	0.69
He et al. [5]	$4T_{hash} + T_{aes} + T_{rand}$	0.23
Proposed	$2T_{puf} + 2T_{hash} + 4T_{rand}$	0.18

Benchmark (avg. 100 iterations): $T_{rand}=0.0018$ ms, $T_{puf}=0.0522$ ms, $T_{hash}=0.0353$ ms, $T_{aes}=0.0867$ ms.

Due to the execution of the authentication and key exchange phase, analysis of its computational efficiency is essential. In the proposed scheme, D and AS (Figures 2 and 3) both execute one PUF operation (T_{puf}).

- D additionally performs six hash operations (T_{hash}): pseudonym computation, unique secret hash, and XOR mask in the Authentication Phase, followed by mask recovery, session key derivation, and pseudonym rotation in the Key Exchange Phase.
- AS performs five hash operations (T_{hash}), along with one AES encryption (T_{aes}) for the gateway token and one random generation (T_{rand}) for m_{new} .
- GW runs one AES decryption (T_{aes}) to recover the session key and updated pseudonym from the forwarded token.

This leads to the total computational cost of the authentication and key exchange phase amounting to $2T_{puf} + 11T_{hash} + 2T_{aes} + 1T_{rand}$. In Table IV, the proposed scheme is said to have a computational cost of 0.67 ms. Its computational efficiency surpasses that of He et al. [5] (1.74 ms), whose ECC scalar multiplication operations incur significantly higher cost on constrained platforms. Due to the incorporation of a

PUF challenge–response pair and an AES-encrypted gateway token — enabling PUF-based authentication and decoupled AS selection — the scheme incurs a modest overhead over LAAKA [19] (0.56 ms) and Zhou [18] (0.53 ms), which neither employ PUF-based authentication, a decoupled distributed AS, nor dual-state desynchronisation recovery.

TABLE IV
COMPUTATIONAL COST COMPARISON: AUTHENTICATION & KEY EXCHANGE PHASE.

Scheme	Operations	Cost (ms)
LAAKA [19]	$16T_{\text{hash}}$	0.56
Zhou [18]	$15T_{\text{hash}}$	0.53
Zheng et al. [9]	$2T_{\text{puf}} + 6T_{\text{hash}} + 4T_{\text{aes}} + 5T_{\text{rand}}$	0.67
He et al. [5]	$11T_{\text{hash}} + 5T_{\text{aes}} + 2T_{\text{ecc}}$	1.74
Proposed	$2T_{\text{puf}} + 11T_{\text{hash}} + 2T_{\text{aes}} + T_{\text{rand}}$	0.67

$T_{\text{ecc}}=0.461$ ms (scaled from He et al. [5] hardware ratios).

2) *Communication Cost*: Communication cost is a core feature for analysing the performance of an authentication scheme, based on the number of message exchanges and their sizes (in bits). By assuming that communications are executed with uniform bit lengths, the communication cost of both enrolment and the authentication and key exchange phase is evaluated and compared with other state-of-the-art schemes.

It is assumed that identities and timestamps are 32 bits, along with nonces, hashes, challenges, responses, and ciphertexts as 256 bits, ensuring a fair comparison.

During the node enrolment phase, D enrolls itself with AS , executing three message exchanges. The first message communicates the unique identity of D , i.e., ID_D (32 bits), to AS over the secure channel. In the second message, AS transmits a challenge c_D (256 bits) and a session-based random m_D (256 bits) to D . The third communication takes place from D to AS in the form of a unique secret y_D (256 bits), a PUF response R_D (256 bits), and a reverse challenge c_{AS-D} (256 bits). Thus, the total communication cost per enrolment is 1312 bits as presented in Table V. It outperforms schemes such as LAAKA [19] (2592 bits) and Zheng et al. [9] (1600 bits), while remaining competitive with Zhou [18] (1344 bits) and He et al. [5] (1280 bits). Unlike those schemes, the proposed scheme binds device enrolment to a PUF-derived secret and initialises the pseudonym at registration, adding security guarantees at a modest communication cost.

TABLE V
COMMUNICATION COST COMPARISON: ENROLMENT PHASE.

Scheme	# Messages	Cost (bits)
LAAKA [19]	3	2592
Zhou [18]	5	1344
Zheng et al. [9]	3	1600
He et al. [5]	2	1280
Proposed	3	1312

IDs/timestamps = 32 b; hashes/nonces/challenges/ciphertexts = 256 b.

TABLE VII
SIMULATION PARAMETERS.

Parameter	Value
Simulator	COOJA (Contiki-NG)
Mote type	Emulated IoT nodes
Network topology (study i)	100 nodes: 1 GW, 79 AS (2 active), 20 devices
Phases measured	Enrolment, Authentication, Key Exchange
Schemes compared	Proposed scheme, LAAKA [19], Zhou [18]
Active AS (study ii)	2, 5, 10
Network size (study iii)	$N \in \{30, 50, 80, 100\}$ nodes
Random seeds	5 per configuration

TABLE VI
COMMUNICATION COST COMPARISON: AUTH. & KEY EXCHANGE PHASE.

Scheme	# Messages	Cost (bits)
LAAKA [19]	3	2400
Zhou [18]	4	2464
Zheng et al. [9]	3	1600
He et al. [5]	3	1120
Proposed	3	1376

The authentication and key exchange phase is executed using three message communications. The first message takes place from D to AS , where D communicates its current pseudonym PID (256 bits), a masked authentication tag Y_{AS-D}^H (256 bits), and a timestamp ts_1 (32 bits). In the second message, AS transmits the masked session seed m_H (256 bits) and a timestamp ts_2 (32 bits) to D . The third communication takes place from AS to GW , consisting of the updated pseudonym PID_{new} (256 bits), the identity ID_{AS} (32 bits), and the encrypted gateway token auth_token_D (256 bits). Thus, the total communication cost becomes 1376 bits, as presented in Table VI. He et al. [5] achieves a lower figure (1120 bits) for a simpler two-party IoT-to-server protocol, whereas the proposed scheme supports multihop mesh networks with a decoupled distributed AS, desynchronisation recovery, and PUF-based authentication — features that account for the additional overhead. DAuth [8] requires 928 bits, with the 448-bit increase in the proposed scheme directly reflecting the widening of the device identifier from a 32-bit ID_D to a 256-bit pseudonym PID , the direct cost of adding device anonymity to all open-channel communications.

B. Simulation-Based Performance Analysis

A simulation-based performance evaluation has been conducted to detect the efficiency of the proposed authentication scheme using the COOJA simulator within the Contiki-NG framework. Three complementary studies are reported. The first is an overall per-device comparison of the proposed scheme against LAAKA [19] and Zhou [18] in a fixed 100-node network comprising 1 gateway, 79 AS nodes (2 active), and 20 newly joined devices. The second is an authentication server (AS) pool-size study that varies the number of active AS nodes from 2 to 10. The third is a network scalability study that grows the total node count across $N \in \{30, 50, 80, 100\}$. All results are averaged over five independent random seeds. Table VII elaborates the common simulation parameters.

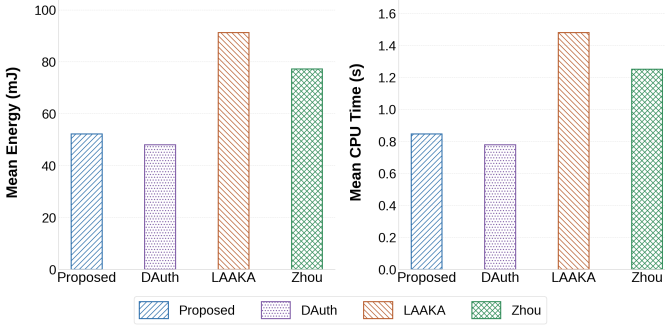


Fig. 5. Per-device mean total cost, all phases: (a) energy and (b) CPU time.

1) *Performance Analysis for Overall Per-Device Cost:* Initially, the authentication schemes are evaluated by considering the overhead of one complete authentication cycle (enrolment, authentication, and key exchange) per device. For 20 newly joined nodes in a fixed 100-node network (1 gateway, 79 AS nodes with 2 active, and 20 newly joined devices), the proposed scheme demonstrated a consistent advantage over LAAKA [19] and Zhou [18] by achieving a reduction of 42.8 % and 34.6 % with respect to total per-device energy consumption (Figure 5a), respectively. Similarly, a reduction of 42.7 % and 35.3 % is demonstrated with respect to total per-device CPU time (Figure 5b) over LAAKA and Zhou, respectively. The dominant savings arise in the authentication and key exchange phase, where the proposed scheme consumes 28.87 mJ against 77.97 mJ for LAAKA and 57.06 mJ for Zhou. Due to the incorporation of fewer message exchanges and computationally lightweight hash-XOR operations, the proposed scheme has minimised radio activity and computational load compared to the message-intensive and cryptographically heavier LAAKA and Zhou schemes.

2) *Performance Analysis for Authentication Server Pool Size:* In the decoupled design, the gateway selects any capable node as the authentication server, so the size of the AS pool directly influences load distribution and overall network performance. The performance of the proposed scheme, LAAKA, and Zhou is evaluated by varying the number of active AS nodes from 2 to 10. Considering the total energy consumption (Figure 6a), a reduction of approximately 42 % is demonstrated by the proposed scheme over LAAKA at every AS count. Both schemes improve from AS=2 to AS=10, with the proposed scheme dropping from 1065 mJ to 936 mJ and LAAKA from 1827 mJ to 1631 mJ, suggesting that a pool of 10 AS nodes is near-optimal for a 20-device network. Similarly, as shown in Figure 6(b), a reduction of approximately 42 % in total CPU time is demonstrated over LAAKA across all AS counts. Zhou's total cost remains flat at approximately 1503 mJ in energy and 24.4 s in CPU time regardless of AS pool size, confirming that its gateway-coupled design is insensitive to AS selection diversity. These consistent reductions demonstrate the energy and computational efficiency of the decoupled AS design in the proposed scheme.

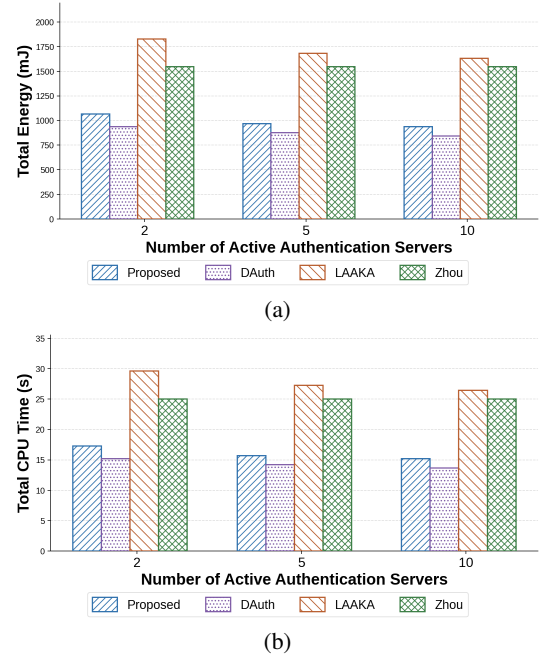


Fig. 6. Total authentication cost vs. number of active AS nodes (20 devices): (a) energy and (b) CPU time.

3) *Performance Analysis for Network Scalability:* The performance of authentication schemes is also analysed by considering their scalability with respect to total network size N , varied from 50 to 100 nodes. Considering the total energy consumption (Figure 7a), a reduction of 42.2 % and 50.6 % is demonstrated by the proposed scheme at $N = 100$ over LAAKA (898 mJ) and Zhou (1051 mJ), respectively. Similarly, as shown in Figure 7(b), a reduction of 42.5 % and 50.6 % is demonstrated with respect to total CPU time over LAAKA (14.6 s) and Zhou (17.0 s), respectively. LAAKA and Zhou exhibit steeper energy and CPU growth with increasing N because their heavier per-device cryptographic load — AES-intensive authentication in LAAKA and a combined heavy authentication and key exchange in Zhou — accumulates faster as the device count grows. Due to the incorporation of lightweight XOR-hash operations and a decoupled AS design, the proposed scheme keeps per-device cost low and achieves the lowest total cost among all compared schemes across all measured network sizes. These consistent reductions across all network sizes demonstrate the scalability and computational efficiency of the proposed scheme in comparison to the benchmark schemes.

4) *Performance Analysis for Desynchronisation Recovery:* During the Key Exchange Phase, the AS simultaneously dispatches two packets: one to the gateway carrying $(PID_{new}, ID_{AS}, auth_token_D)$, and one to the device carrying (m_H, ts_2) . Desynchronisation arises when the packet destined for the device is lost in transit, for any reason. After such a loss, the device retains its old state m_{curr} while the AS has already rotated to a new state, causing subsequent authentication attempts to fail.

DAuth [8] stores only a single nonce state at the AS. When the device re-authenticates using the old m_{curr} , the AS has no means to recognise it and rejects the request outright. The device must therefore undergo a full re-enrolment (Phase 1: PUF challenge-response and accumulator setup) before a new authentication and key exchange can succeed. The proposed scheme eliminates this penalty through its dual-state mechanism: the AS retains both $(m_{\text{curr}}, \text{PID}_{\text{curr}})$ and $(m_{\text{old}}, \text{PID}_{\text{old}})$. When the device arrives with the old pseudonym, the AS recognises it via the PID_{old} lookup and runs the full authentication protocol: recovering the PUF response R_D , deriving $Y_{\text{AS-D}}^H$ from the device's unique secret y_D , and verifying the accumulator using m_{old} , then completing a fresh key exchange, all without re-enrolment.

Figure 8 presents a per-device energy and CPU time comparison of a single authentication round under two conditions: normal operation (before any packet loss) and recovery (the round immediately following a desynchronisation event), evaluated over a 100-node multi-hop RPL network, 20 devices, and 5 independent seeds. As shown in Figure 8(a), DAuth [8] incurs 134.8 mJ per recovery round, a 118.3% increase over its normal round cost of 61.8 mJ, reflecting the overhead of a failed authentication attempt followed by full re-enrolment. The proposed scheme's recovery round costs 87.2 mJ, statistically indistinguishable from its normal round (98.6 mJ), confirming that the dual-state mechanism incurs zero additional overhead during recovery. Compared directly, the proposed scheme reduces per-device recovery energy by **47.6 mJ (35.3 %)** over DAuth [8]. The CPU time results in Figure 8(b) exhibit the same pattern: DAuth [8] incurs 2.185 s per recovery round against 1.413 s for the proposed scheme, yielding a **35.3 % reduction** in execution time.

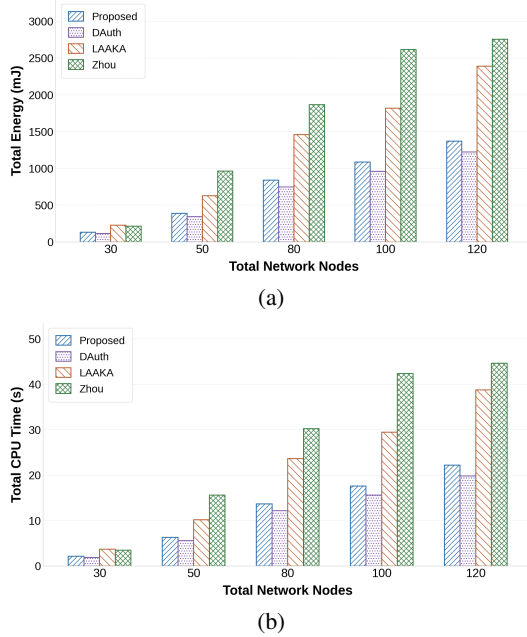


Fig. 7. Network scalability: (a) total energy and (b) total CPU time vs. network size N .

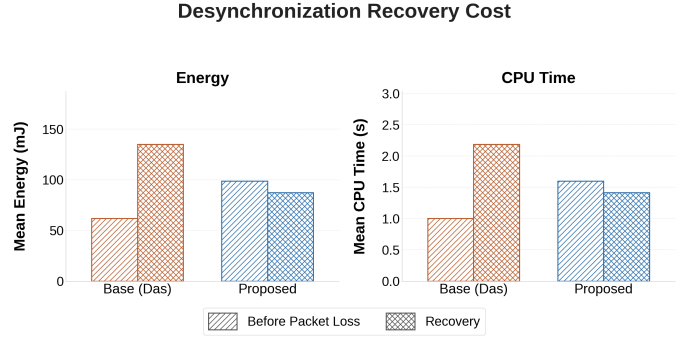


Fig. 8. Per-device desynchronisation recovery cost: (a) energy and (b) CPU time for a normal round vs. a recovery round, averaged over 20 devices and 5 seeds in a 100-node RPL network.

C. Hardware-Based Performance Analysis

Hardware-based performance analysis plays a crucial role as it validates the feasibility of the proposed authentication scheme by implementing it on physical IoT devices, incorporating real hardware environment conditions. The proposed scheme and the three comparison schemes—DAuth [8], LAACA [19], and Zhou [18]—are implemented on a Raspberry Pi 4B platform, equipped with a quad-core 64-bit ARM Cortex-A72 processor operating at 1.5 GHz. A uniform hardware setup is maintained across all compared authentication schemes, ensuring a fair and consistent comparison of energy consumption and CPU time. The implementation considers two Raspberry Pi 4B platforms as IoT nodes. One performs the role of the newly joined device, initiating the authentication process. The other functions as the authentication server, performing the necessary verification for authentication and session key exchange. A Linux-based laptop is leveraged as the gateway, handling authentication verification and session key delivery. Each of the three entities maintains direct TCP communication with the others. Energy consumption is estimated using the active power profile of the Raspberry Pi 4B (3.8 W) multiplied by the measured wall-clock time per authentication round, capturing both cryptographic computation and network round-trip latency.

As illustrated in Figure 9(a), DAuth achieves the lowest per-round energy of 0.152 J, as it performs no pseudonym operations. The proposed scheme consumes 0.286 J per round—an overhead of approximately 88% over DAuth attributable to the additional SHA-256 evaluations for PID rotation and dual-state verification. Despite this overhead, the proposed scheme remains 14.2% and 52.5% more energy-efficient than Zhou (0.333 J) and LAACA (0.602 J), respectively, confirming that the anonymity guarantee is achieved at a cost well within the range of existing non-anonymous schemes.

The CPU time results in Figure 9(b) follow the same ordering: DAuth (0.040 s), Proposed (0.075 s), Zhou (0.088 s), and LAACA (0.159 s). The proposed scheme reduces CPU time by 14.8% and 52.8% relative to Zhou and LAACA, respectively, while incurring an 87.5% overhead compared to DAuth—consistent with the energy results. These reductions

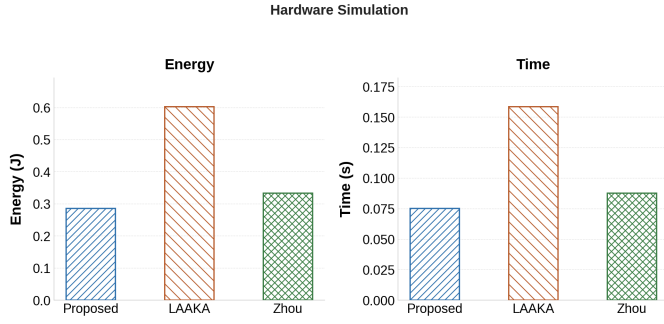


Fig. 9. Hardware-based performance: per-round energy (a) and CPU time (b) on Raspberry Pi 4B (Proposed, DAAuth, LAAKA, Zhou). DAAuth is the base scheme with no pseudonym operations; the proposed scheme adds $\approx 88\%$ overhead over DAAuth for anonymity while remaining 52.5% and 14.2% cheaper than LAAKA and Zhou, respectively.

are consistent with the lightweight XOR-hash operations and decoupled AS design of the proposed scheme, which translate directly to measurable gains in physical energy and execution time on real hardware. The inconsistency between hardware and simulation-based results arises due to the disparity in scale, where a 100-node network is assumed in simulation whereas the hardware setup involves only two Raspberry Pi platforms.

VI. CONCLUSION

This paper presented a lightweight, anonymous, and decoupled distributed authentication scheme for multihop IoT networks. The assigned authentication server, which may or may not be the parent of the newly joined device, performs authentication (Authentication Phase) and session key establishment (Key Exchange Phase) in sequential phases. Device anonymity is preserved through per-session pseudonym rotation, preventing adversarial tracking across sessions. A dual-state mechanism at both the authentication server and the gateway provides transparent desynchronisation recovery without re-enrolment. PUF-based authentication eliminates long-term secret storage, and hash-based accumulator operations with XOR masking keep the computational overhead low.

Formal verification in ProVerif confirms authentication correspondence, session key secrecy, resistance to offline guessing attacks, and observational-equivalence-based anonymity. COOJA/Contiki-NG simulations confirm practical suitability. The proposed scheme achieves a per-device three-phase energy of 52.21 mJ, reducing total energy by 42.8% over LAAKA [19] and 34.6% over Zhou [18], with consistent gains across varying network sizes (up to $N = 100$ nodes) and AS pool sizes.

REFERENCES

- [1] I. Zhou, I. Makhdoom, N. Shariati, M. A. Raza, R. Keshavarz, J. Lipman, M. Abolhasan, and A. Jamalipour, "Internet of things 2.0: Concepts, applications, and future directions," *IEEE Access*, vol. 9, pp. 70961–71012, 2021.
- [2] N. Lajnef, A. Mami, and F. Derbel, "Applications of wireless sensor networks and IoT frameworks in industry 4.0: A systematic literature review," *Sensors*, vol. 22, no. 6, 2022.
- [3] M. Khatua, S. Das, and P. M. Rana, "PUF-based lightweight mutual authentication scheme for IoT-based smart grids," in *Proc. IEEE ANTS*. IEEE, 2024, pp. 1–6.
- [4] S. A. Sheik and A. P. Muniyandi, "Secure authentication schemes in cloud computing with glimpse of artificial neural networks: A review," *Cyber Security and Applications*, vol. 1, p. 100002, 2023.
- [5] D. He, Y. Cai, S. Zhu, Z. Zhao, S. Chan, and M. Guizani, "A lightweight authentication and key exchange protocol with anonymity for IoT," *IEEE Trans. Wireless Communications*, vol. 22, no. 11, pp. 7862–7872, 2023.
- [6] S. Roy, M. H. Mahalat, and B. Sen, "Design and implementation of cost-effective end-to-end authentication protocol for PUF-enabled IoT devices," *IEEE Trans. Emerging Topics in Computing*, 2025.
- [7] P. Rosa, A. Souto, and J. Cecilio, "Light-SAE: A lightweight authentication protocol for large-scale IoT environments made with constrained devices," *IEEE Trans. Network and Service Management*, vol. 20, no. 3, pp. 2428–2441, 2023.
- [8] S. Das and M. Khatua, "Lightweight and decoupled distributed authentication for multihop IoT networks," in *Proc. IEEE COMSNETS*, 2026.
- [9] Y. Zheng, W. Liu, C. Gu, and C.-H. Chang, "PUF-based mutual authentication and key exchange protocol for peer-to-peer IoT applications," *IEEE Trans. Dependable and Secure Computing*, vol. 20, no. 4, pp. 3299–3316, 2022.
- [10] O. Alruwaili, M. Tanveer, S. A. Aldossari, S. Alanazi, and A. Armghan, "RAAF-MEC: Reliable and anonymous authentication framework for IoT-enabled mobile edge computing environment," *Internet of Things*, vol. 29, p. 101459, 2025.
- [11] S. Li, T. Zhang, B. Yu, and K. He, "A provably secure and practical PUF-based end-to-end mutual authentication and key exchange protocol for IoT," *IEEE Sensors Journal*, vol. 21, no. 4, pp. 5487–5501, 2020.
- [12] W. Yang, C. Hou, Y. Wang, Z. Zhang, X. Wang, and Y. Cao, "SAKMS: A secure authentication and key management scheme for IETF 6TiSCH industrial wireless networks based on improved elliptic-curve cryptography," *IEEE Trans. Network Science and Engineering*, vol. 11, no. 3, pp. 3174–3188, 2024.
- [13] S. Al-Riyami and K. G. Paterson, "Certificateless public key cryptography," in *Proc. ASIACRYPT*, ser. LNCS, vol. 2894. Springer, 2003, pp. 452–473.
- [14] M. T. Al Ahmed, F. Hashim, S. J. Hashim, and A. Abdullah, "Authentication-chains: Blockchain-inspired lightweight authentication protocol for IoT networks," *Electronics*, vol. 12, no. 4, p. 867, 2023.
- [15] J. Zhong, S. He, Z. Liu, and L. Xiong, "Lightweight anonymous authentication for IoT: A taxonomy and survey of security frameworks," *Sensors*, vol. 25, no. 17, p. 5594, 2025.
- [16] B. Blanchet, B. Smyth, V. Cheval, and M. Sylvestre, "ProVerif 2.00: Automatic cryptographic protocol verifier, user manual and tutorial," INRIA, Tech. Rep., 2018.
- [17] D. Dolev and A. Yao, "On the security of public key protocols," *IEEE Trans. Information Theory*, vol. 29, no. 2, pp. 198–208, 1983.
- [18] X. Zhou, S. Wang, K. Wen, B. Hu, X. Tan, and Q. Xie, "Security-enhanced lightweight and anonymity-preserving user authentication scheme for IoT-based healthcare," *IEEE Internet of Things Journal*, vol. 11, no. 6, pp. 9718–9731, 2024.
- [19] H. Ali and I. Ahmed, "LAAKA: A lightweight anonymous authentication and key agreement scheme for IoT-fog environment," *Computers & Security*, vol. 140, p. 103770, 2024.